Evidence-Bounded Retrodiction

How an AI Can Reason About the Past Without Giving the Past Knowledge of the Future

A Plain-English Explanation, Mathematical Formalization, Worked Examples, and Open Problems

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Technical companion to: From Memory to Causal History: Evidence-Bounded Retrodiction for Persistent Agents

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Central idea: Later evidence may change our understanding of the past. It must not change what was actually knowable in the past.

1. The problem in plain English

Imagine judging a decision made ten years ago. Today, we know how everything turned out. We know which warning signs mattered, which assumptions failed, and which later discoveries changed the story. The person making the decision ten years ago did not.

If an AI uses everything known today to reconstruct what that person 'should have known' then, it commits a subtle hindsight error: it gives the past information from the future.

Evidence-Bounded Retrodiction (EBR) is a protocol for preventing that mistake while still allowing later evidence to improve our reconstruction of what happened. The published paper frames this as a three-view temporal-epistemic protocol for persistent agents: contemporaneously attested actor state, later reconstruction of historical actor access, and retrospective causal evidence.

The aim is not to make an AI certain about history. The aim is to make it disciplined about which evidence belongs to which historical question.

2. Three questions that look similar but are not

At a historical decision time, three questions must be kept separate:

Question	Plain-English meaning	EBR view
What was documented as available then?	What can we contemporaneously attest was in the actor's information state at the decision time?	K_att
What can we later prove the actor had access to then?	Later provenance may reveal historical access that the original archive did not yet	K_rec

	contain.	
What do we know now about what happened?	Later inspections, outcomes, archives and measurements may help reconstruct the causal history.	R(c)

These sets may overlap. They are not interchangeable.

3. Three metaphors

3.1 The detective and the driver

A driver chooses a route at 9:58. A crash occurs at 10:00. At noon, investigators discover a mechanical defect. The noon inspection may be excellent evidence about why the crash happened. It is not evidence that the driver knew about the defect at 9:58.

Now suppose a warning message was actually seen by the driver at 9:57, but the server archive containing that message was imported only at 11:30. Database record time alone would wrongly exclude something the driver really had access to.

EBR needs both distinctions at once: later evidence can illuminate the causal story, and later provenance can illuminate historical access, without rewriting the contemporaneous record.

3.2 The government decision

Suppose a public authority makes a flood-control decision in 2022. A 2025 engineering study later reveals a drainage failure that was physically present in 2022 but not known to the decision-makers. The study can inform retrospective causal analysis. It cannot fairly be used as though officials possessed it in 2022.

Conversely, if meeting minutes discovered in 2025 prove that officials were shown a warning in 2022, that later discovery can update our reconstruction of what they had access to then. It still does not mutate the original 2022 record.

3.3 The persistent AI agent

An AI assistant recommends a supplier in January. In March the supplier fails. In April the AI learns facts that were unavailable in January. A naive memory system may later explain the January recommendation using April knowledge and conclude that the recommendation was obviously bad.

EBR instead asks two separate questions: Was the January decision reasonable given the January evidence? And, using everything learned by April, what causal history best explains the supplier failure? Learning from the outcome should influence future reasoning, but it should not falsify the historical information state.

4. Why ordinary timestamps are insufficient

An evidence item can have several different temporal meanings. EBR explicitly separates them.

Symbol / concept	Meaning	Why it matters
$I_v(x)$	Event or valid interval for evidence item x	When the underlying event/state belongs in world time.
$r(x)$	System record time	When the system stored or imported x.
$\pi(x)$	Provenance	Source and lineage of x.
$\sigma(x)$	Epistemic status	Whether x is documented, inferred, speculative, contradicted, unknown, etc.
$A(a,x,t)$	Actor-access assertion	Provenance-bearing evidence that actor a had access to x by time t.

Actor access cannot safely be inferred from event time: an event may occur without an actor knowing it. Nor can it be inferred from record time: an actor may see something before a later system import.

5. The mathematical core

Let a be an actor, t_i a historical decision time, and c an audit cutoff with $c \geq t_i$. Let $C(a,x,t_i)=1$ mean that access to x by t_i was contemporaneously attested by t_i. Let $H_c(a,x,t_i)=1$ mean that evidence available by audit cutoff c supports historical access to x by t_i.

5.1 Contemporaneously attested state

$$K_{att}(a,t_i) = \{ x : r(x) \leq t_i \text{ and } C(a,x,t_i) = 1 \}$$

What it means: K_{att} contains evidence that was already recorded by the decision time and contemporaneously attested as accessible to the actor.

Why we need it: this is the strongest protected historical snapshot for ex-ante appraisal.

What it does not prove: it is not a complete model of everything inside the actor's mind. It is an evidence-bounded representation of contemporaneously attested access.

5.2 Reconstructed historical access

$$K_{rec}(a,t_i;c) = \{ x : r(x) \leq c \text{ and } H_c(a,x,t_i) = 1 \}$$

What it means: using evidence available by later cutoff c, reconstruct what the actor had access to by the old decision time t_i.

Why we need it: archives can arrive late. A message seen before a decision may only enter the system afterward.

What it does not prove: inclusion says later evidence supports historical access. It does not say the content of x was true.

5.3 Retrospective evidence

$$R(c) = \{ x : r(x) \leq c \} \cap S$$

Here S is the declared retrieval scope.

What it means: R(c) contains evidence recorded by audit cutoff c and inside the declared scope, whether or not the historical actor had access to it.

Why we need it: retrodiction is allowed to use later inspections, outcomes and archives to understand what happened.

What it does not permit: R(c) must not be silently substituted for a historical actor-state snapshot when judging what was reasonable at t_i.

6. The non-retroactivity invariant

The core architectural rule assumes append-only transaction history: later records and later access attestations cannot be back-written as if they existed earlier.

If a record is inserted after t_i, then K_att(a,t_i) is invariant.

Later evidence may expand K_rec(a,t_i;c) or R(c) as c advances, but it cannot mutate K_att(a,t_i). Corrections are new records linked to superseded assertions rather than destructive edits.

> **The detective may learn more. The detective may even learn what the historical actor had seen. The detective may not edit the old evidence ledger to pretend the later discovery was recorded there all along.**

This invariant is an application of established versioned/bitemporal semantics, not a claim of a new database theorem.

7. Reconstructing candidate causal histories

EBR does not assume that temporal sequence proves causation. A candidate history is generated from an observed present state, a declared scope, facts, assumptions, candidate hypotheses, exclusions and potentially missing variables or sources.

A practical generator can retrieve evidence, propose candidate causes, expand required mechanism bridges, search for contradictions, deduplicate common source lineages, merge related branches and prune the search space.

For ex-ante appraisal, candidate generation should be blind to retrospective outcome-conditioned evidence on its first pass. Retrospective evidence enters separately. This makes hypothesis-space leakage auditable.

8. The mechanism gate

A proposed causal edge u → v is not admitted merely because u happened before v or resembles prior cases. A required mechanism bridge should carry a named mechanism class, temporal status, supporting provenance roots, an expected or testable intermediate implication, contradiction status and an epistemic label.

Similarity can nominate a candidate; causal admissibility requires mechanism + evidence.

If a required bridge is unresolved, the history can remain an investigatory hypothesis but is not called a complete admissible explanation. Where a valid structural causal model exists, interventions and counterfactuals can be evaluated under that model. Without such assumptions, EBR does not manufacture causal effect estimates.

9. Evidence profiles without pseudo-probability

$$\Phi(h) = \langle\, m(h),\ n_{\pi}(h),\ c(h),\ u(h) \,\rangle$$

For candidate history h:

- m(h): supported / required mechanism-bridge coverage.
- $n_{\pi}(h)$: independent provenance roots after source-lineage deduplication.
- c(h): burden of adjudicated contradictions under the domain evidence policy.
- u(h): unresolved required mechanism bridges.

The profile is deliberately non-probabilistic. The published prototype avoids collapsing these dimensions into an unvalidated weighted scalar, because correlated evidence can be double-counted, weights are domain-dependent, and a closed candidate set may be incomplete.

The prototype uses a transparent non-compensatory display ordering: complete histories with no adjudicated hard contradiction precede incomplete or hard-contradicted histories. Ties or incomparable candidates may remain.

A stronger mathematical formulation may replace this display convention with a partial order or dominance relation; that is an open problem rather than an established result.

10. Open-set uncertainty

$$\perp = \text{unmodeled-history reserve}$$

Suppose the system examines H1, H2 and H3. It should not assume one of them must be the truth merely because those are the histories it happened to enumerate.

The reserve ⊥ represents the possibility that the true or best explanation is outside the modeled candidate set. EBR therefore permits language such as 'strongest among the enumerated admissible candidates' and permits complete abstention.

⊥ is not assigned an arbitrary percentage in the current formulation. It marks non-exhaustiveness; it does not solve it.

11. Worked synthetic example

The published prototype uses an automated production cell. Power is lost at 10:01, an operator makes a response decision at 10:05, and an audit occurs at 12:00.

- A grid warning was seen by the operator at 09:57, but its message archive entered the system only at 11:30.
- An 11:20 inspection reports damage pertaining to a 09:56 disturbance. That inspection was not available to the operator at 10:05.
- Three candidate histories are enumerated: external grid disturbance, internal overload, and manual shutdown.

Historical view	Leakage	Actor precision	Actor recall
Valid-time-only	1	.83	1.00
Record-time-only	0	1.00	.80
Three-view K_rec	0	1.00	1.00

Valid-time-only leaks the later inspection into the actor's historical state because the inspection concerns an earlier event. Record-time-only misses the grid warning because its archive arrived late. The three-view reconstruction handles both timing cases correctly in this deterministic toy world.

At audit time, the external-disturbance history has all three required mechanism bridges supported by four independent provenance roots and no contradiction. The overload history has an unresolved bridge. The manual-shutdown history has supporting chat evidence but is contradicted by an authentication log. The prototype reports the external-disturbance history as strongest among the enumerated admissible candidates, keeps $\perp$ active, and recommends an independent telemetry check.

This demonstrates implementation of the protocol contract only. It does not establish real-world causal accuracy.

12. How the protocol can be falsified

The published research proposes tests that can fail. That is important: EBR should earn its complexity empirically.

- Temporal leakage: does post-cutoff information contaminate ex-ante appraisal?
- Actor-state fidelity: when ground-truth access logs exist, how accurately is historical access reconstructed?
- Causal path / bridge recovery: where a ground-truth causal structure exists, are required paths recovered?
- Contradiction recall and provenance-root correctness: are material conflicts and common source lineages preserved?
- Selective risk / abstention: does the system abstain when decisive evidence or the true history is withheld?
- Historical-state invariance: does later evidence leave K_att unchanged?

The core falsifiable question from the paper is whether the three-view contract reduces historical-state errors and hindsight contamination relative to simpler temporal-memory baselines without degrading causal recovery or abstention behavior.

13. What EBR does not claim

- It does not claim that chronology alone proves causation.
- It does not claim that $\Phi(h)$ is a probability distribution.
- It does not claim a new general theory of causality, temporal databases, provenance or abduction.
- It does not claim that later actor-access reconstruction is mind-reading.
- It does not claim that $\perp$ can recover evidence that does not exist.
- It does not claim real-world causal validity from a deterministic toy prototype.
- It does not eliminate post-cutoff knowledge already embedded parametrically in an underlying language model; that is a separate leakage problem requiring separate controls.

14. What I think I have - and what I am asking mathematics to determine

EBR currently contains a mixture of architectural requirements, mathematical definitions, invariants, ordering rules and research hypotheses. I do not claim that all of these constitute novel mathematics.

The research question is whether the underlying requirements admit a stronger and more standard formal structure - and, if they do, what mathematical framework expresses them most faithfully.

Area	Question for formal review
Temporal-epistemic state	Should K_att, K_rec and R be represented as sets, filtrations, indexed evidence structures, lattices, or another standard object?
Non-retroactivity	Can the invariant be stated as a precise proposition with minimal assumptions? Which monotonicity properties should K_rec and R satisfy as c increases?
Actor access	What is the cleanest formal semantics for provenance-bearing historical access assertions without conflating access with belief or truth?
Provenance dependence	How should evidence roots be compared without casually assuming statistical independence?
Contradictions	Should contradiction preservation use paraconsistent logic, signed evidence, constraints, lattices, or domain-specific adjudication?
Mechanism bridges	Can admissibility be expressed as graph

	constraints, proof obligations, or another formal structure?
Candidate ordering	Would a partial order / Pareto dominance relation better preserve incomparability than the prototype's lexicographic display rule?
Open-set reserve	What is the cleanest semantics for ⊥ when the hypothesis set is explicitly non-exhaustive but unsupported probabilities are prohibited?
Probabilistic extension	If a domain supplies calibrated likelihoods or a valid structural causal model, how should probabilities be added without collapsing the three-view separation?
Evaluation	Which formal properties can be tested independently of causal accuracy: invariance, leakage prevention, monotonicity, contradiction preservation and abstention consistency?

15. A possible stronger formal direction

One possible direction, subject to mathematical review, is to represent contemporaneous actor information and retrospective audit evidence as distinct time-indexed provenance-bearing processes. Later audit evidence may update beliefs about historical access and causal history but may not mutate the contemporaneously attested process.

Candidate histories could then inhabit a constrained causal-graph space, with admissibility defined through mechanism obligations and contradiction policy, and comparison defined by a partial order that preserves incomparability. Open-set uncertainty would remain explicit unless a domain-specific probabilistic model justifies calibrated posterior claims.

This is a proposed direction for formalization, not a theorem or an established result of the published paper.

16. Why this matters beyond the toy example

A persistent agent that learns from decisions needs to distinguish process quality from outcome quality. A sound decision can produce a bad outcome because of an unforeseeable shock. A poor process can produce a lucky success. If later outcome knowledge is allowed to rewrite the historical information state, the agent may learn exactly the wrong lesson.

The same distinction can matter in industrial incidents, aviation, cyber investigations, supply chains, organizational memory, historical analysis and scientific hypothesis generation, but each domain requires its own evidence standards, mechanism models and expert validation.

17. The larger objective

The long-term objective is not an AI that is certain about the past. It is an AI that is explicit about the boundary between historical knowledge and retrospective knowledge.

A mature system should be able to say both:

> **Given what was supportably available then, this decision was or was not justified. | Given what we know now, this is the strongest evidence-bounded reconstruction among the histories examined.**

Those are different claims. EBR is designed to stop one from silently becoming the other.

If the protocol succeeds, a persistent agent can use later outcomes to improve future reasoning while preserving the integrity of the historical record.

In the simplest terms: allow an AI to learn from history without rewriting history.

18. Relationship to the published paper

This document is a companion explanation and formalization aid. The canonical research paper is: Ricardo Gill, “From Memory to Causal History: Evidence-Bounded Retrodiction for Persistent Agents,” 2026, Zenodo, DOI 10.5281/zenodo.22681856.

Where this companion proposes stronger mathematical directions or asks open questions, those are explicitly prospective and should not be read as claims already established by the canonical paper.

Declaration on Generative AI

Generative AI tools were used to support drafting, restructuring, mathematical/notation checking, document production and adversarial review. Ricardo Gill directed the conceptual requirements and remains responsible for the claims and final document. The canonical research paper contains the literature review and primary citations supporting the research context.